FastSpeech 2 for English and Telugu

Technical report: implementation, data, training and analysis of two separately trained FastSpeech 2 models (English: Svarah; Telugu: IndicTTS Telugu)
Code, results and checkpoints: `https://github.com/ashdev-7/iiith-fastspeech2` · November 2026

Sources of numbers. Every quantitative claim is labelled: paper reported in the FastSpeech 2 paper (Ren et al., ICLR 2021; LJSpeech); repo taken from the reference implementation `ming024/FastSpeech2` (commit d4e79eb) or its configs; measured measured in this project (Kaggle jobs, logged in each job's `status.json`); estimated an estimate or interpretation, not a measurement. Perceptual statements about the audio are based on automatic transcription and on the predicted/measured variance values; no listening test (MOS) was run.

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Results at a glance

	English (Svarah)	Telugu (IndicTTS)
Training data used (train / validation)	4.87 h / 0.98 h, 110 / 109 speakers measured	4.96 h / 1.02 h, 2 speakers measured
Training	17,949 steps, 3.5 h on 1 × T4	55,524 steps, 8 h on 2 × T4
Validation loss (total), best → final	2.651 (7k) → 2.686	1.970 (16k) → 1.974
Validation duration MAE / Pearson r (log)	2.81 frames / 0.58	1.06 frames / 0.85
Validation pitch RMSE / r (phone level)	34.6 Hz / 0.35	29.2 Hz / 0.76
Whisper-large-v3 CER on validation synthesis (copy-synthesis ceiling)	15.7 % (11.6 %)	35.8 % (33.1 %)
Real-time factor (acoustic + vocoder), T4 / 4-thread CPU	0.027 / 0.35	0.025 / 0.38
Required outputs	10 wavs, 30 variance plots, training and validation loss plots: all produced; every test sentence passes the run-time checks (Σ durations = mel length, no zero-length phones, finite non-silent audio) measured	

1. Implementation approach

1.1 Architecture used from the reference implementation

The starting point is `ming024/FastSpeech2` (MIT licence), vendored unchanged except for the modifications in §1.3. The model is the non-autoregressive FastSpeech 2 acoustic model:

- **Encoder**: phoneme embedding + 4 feed-forward Transformer (FFT) blocks (self-attention, 2 heads, hidden 256; Conv1D 1024, kernels 9/1).
- **Speaker embedding** (multi-speaker mode) added to the encoder output: 110+ Svarah speakers, 2 Telugu speakers.

- **Variance adaptor**: three predictors (2 × Conv1D 256, kernel 3, LayerNorm, dropout 0.5) for log-duration, pitch and energy. Pitch and energy are predicted *per phone*, quantised into 256 bins, embedded and added to the hidden sequence; the **length regulator** repeats each phone vector d times, with d = round(exp(log $\hat{d}$) − 1) at inference.
- **Decoder**: 6 FFT blocks → linear to 80 mel bins → 5-layer convolutional **PostNet** (residual).
- **Loss**: MAE on mel (before and after PostNet) + MSE on log-duration, pitch and energy (6 terms logged).
- **Vocoder**: the reference repository's pretrained **HiFi-GAN "universal"** generator (not trained here).

Aspect	Paper (LJSpeech) paper	Reference repo repo	This project
Data	LJSpeech, 24 h, 1 speaker	LJSpeech / AISHELL-3 / LibriTTS configs	~5 h per language; 110 / 2 speakers
Pitch target	frame-level F0 via continuous wavelet transform	phone-level average F0 (FastPitch-style), no CWT	as repo
Decoder layers / dropout	4 / 0.1	6 / 0.2	as repo
Batch / steps	48 / 160k	16 / 900k (LJSpeech config)	16 / 18k (en), 56k (te), sized from measured throughput
Vocoder	Parallel WaveGAN	HiFi-GAN / MelGAN	HiFi-GAN universal
Reported quality / speed	MOS 3.83 ± 0.08; RTF 1.95×10^{-2} on a V100 (incl. vocoder)	—	RTF 2.5–2.7 × 10^{-2} on a T4 measured; no MOS

The paper's MOS and RTF cannot be compared directly with ours: different data (24 h single speaker vs 5 h), vocoder, GPU, and no listening test here.

1.2 Language-specific work

Telugu (no Telugu support exists in the reference)

- **Grapheme-to-phone** (`text/telugu/g2p.py`, `phones.py`): rule-based conversion of Telugu script, which is close to phonemic: consonants carry the inherent vowel /a/ unless followed by a vowel sign or virama; the anusvara (ం) becomes the nasal homorganic with the next consonant (e.g. బలంగా → `b a l a ng g aa`, విశ్లేషించారు → `… ny c aa r u`), otherwise /m/; visarga → /h/. 50 phone labels in ASCII (retroflex consonants doubled: `tt dd nn ss ll`; aspirates with `h`; long vowels doubled). Every code point must be consumed by a rule, otherwise an error is raised, so characters are never silently dropped.
- **Normalisation** (`text/telugu/normalize.py`): Telugu number words for cardinals, ordinals and times, e.g. 2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు.
- **Symbol table**: a separate 65-symbol Telugu table (padding, punctuation, 50 phones, silences); the encoder embedding is sized per language (L1–L4 in `MODIFICATIONS.md`).
- **Alignment**: no pretrained Telugu MFA model exists, so an MFA acoustic model was **trained from scratch** on the selected 6 h with the rule-based dictionary (11,947 words, 0 without pronunciation) measured.
- **Code-mixing** (beyond the test set): English words written in Latin script inside Telugu text (e.g. "iPhone") are mapped from English ARPAbet to the nearest Telugu phones, using only phones already in the Telugu table (`text/telugu/latin.py`).

English

- **Inference frontend** (`text/english_frontend.py`): times ("10:45 AM" → "ten forty five A M", with AM/PM as letter names), numbers and years (reference `inflect` rules: 2026 → "twenty twenty six"); homographs known to g2p_en (e.g. *conduct*) resolved by part of speech over the whole sentence; first pronunciation from the LibriSpeech lexicon (the lexicon used for alignment); g2p_en for out-of-vocabulary words; punctuation → short pause `sp`.
- **Two rules added after the first full run** (§5.1): split words ("rhyth m") are joined for pronunciation when the concatenation is a lexicon word and the first part is not; a word-final "oo" in an out-of-vocabulary word is /uː/ ("andaroo" → … R UW). The test text itself is never changed; every decision is recorded in `frontend_trace.json`.

- **Alignment**: MFA's pretrained `english_us_arpa` acoustic model with the LibriSpeech lexicon + g2p_en for out-of-vocabulary words (6,388 words, 0 without pronunciation) measured.

1.3 Modifications to the reference code

Twelve fixes (M1–M12) and four language-support changes (L1–L4) are listed in `MODIFICATIONS.md` with the original behaviour, the reason, the new behaviour and the test that covers it. The ones that change results:

#	Problem in the reference	Fix
M1	Energy embedding received the *pitch* control value	uses `e_control`
M5	Utterances without a TextGrid added the previous utterance's pitch/energy to the normalisation statistics	statistics only from processed utterances
M6	Empty TextGrid intervals (MFA ≥ 2 writes pauses this way) were dropped, so pauses vanished from phone sequences and durations	kept as `sp`
M9	float64 energy targets: `backward()` failed ("Found dtype Double")	cast to float32
M10	numpy-scalar learning rate in checkpoints, rejected by `torch.load(weights_only=True)`	`float()`
M12	Pitch/energy controls multiplied the *z-normalised* value: for a voice below the corpus mean, pitch × 1.2 *lowered* F0 (126 → 113 Hz) measured	scale in Hz / energy units; now × 1.2 → 126 → 157 Hz (en), 209 → 252 Hz (te) measured

Others: CPU-safe STFT (M2), librosa ≥ 0.10 keyword API (M3), UTF-8 file handling (M4, M7), sorted speaker ids (M7), `map_location`/`weights_only` loading and HiFi-GAN paths (M8), masks on the input's device (M11).

1.4 Engineering for a reliable run

All compute ran as Kaggle script jobs that clone the repository and record the commit. Each job first runs the test suite (75 tests on Kaggle: frontends, symbol tables, split leakage, preprocessing, model forward/backward on every test sentence, checkpoint round trip, plots); a CPU **rehearsal** ran every stage on a small real subset before the full runs. The training job measures throughput for 300 steps, sizes the schedule to a time budget, and runs a **dry run of inference and plotting** on that 300-step checkpoint before the long training. A **watchdog** kills a training process that exceeds its budget or stops logging; this was added after the first English run hung inside training (two-GPU DataParallel) and was killed at Kaggle's 12-hour limit with no checkpoint. The rerun used one GPU and completed.

2. Data: selection, split and preprocessing

2.1 Selection and subsampling

Both datasets are larger than the 5 h + 1 h budget, so a deterministic, duration-targeted selection was used (`data/select_split.py`):

- **Unit of assignment = group**, never a single clip: a Svarah *recording* (all chunks of one recording), or for Telugu all utterances with the same normalised transcript (both IndicTTS speakers often read the same sentence). Groups that share a transcript are additionally merged (union-find), so no sentence can occur in both splits.
- **Order**: groups are visited in a seeded pseudo-random order (SHA-1 of seed and group id), independent of file order.
- **Quotas**: each speaker gets train/val hours proportional to its share of the candidate audio; a group goes to validation if it fits all validation quotas it touches, else to training if it fits, else it is not used.
- **Leakage checks**: utterance ids, groups, normalised transcripts and audio hashes (PCM SHA-1) must be disjoint between train and validation; duplicate audio is excluded. All checks passed (0 overlaps) measured.

	English (Svarah)	Telugu (IndicTTS)

Selected train	3,079 utt. · 4.87 h · 110 speakers	2,164 utt. · 4.96 h (female 2.41 h, male 2.55 h)
Selected validation	742 utt. · 0.98 h · 109 speakers	450 utt. · 1.02 h
Not aligned by MFA	0	28 (1.1 %), in 4 clusters of consecutive ids
Rejected in preprocessing	2 (≤ 1 voiced frame)	0
Used for training / validation	3,077 / 742	2,140 / 446

The unaligned Telugu clips are clustered (ids te_0091x–0092x, te_0233x, te_0264x), which suggests transcript/audio mismatches in those ranges rather than random failures estimated; they were excluded, not repaired. Svarah validation speakers also appear in training (different recordings): the split is by recording, not by speaker, so the validation set measures generalisation to new sentences, not to new voices.

2.2 Preprocessing

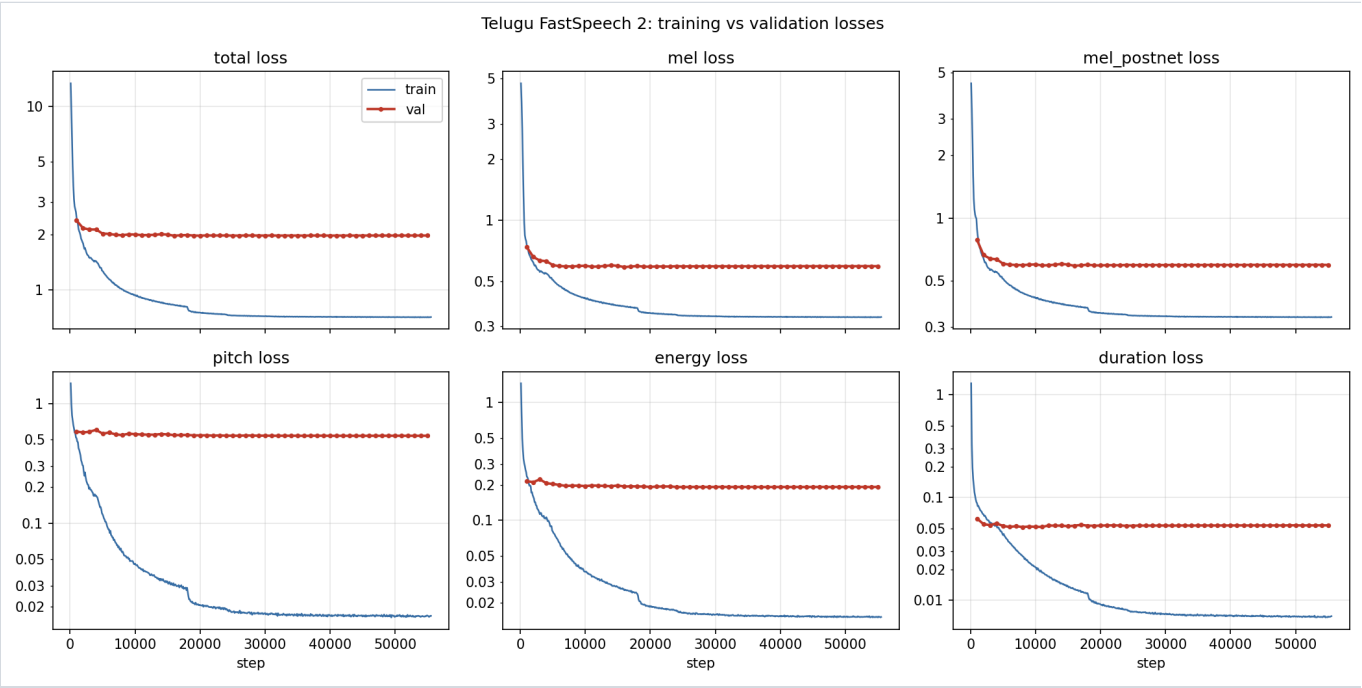
1. **Audio**: decoded from the Hugging Face parquet files, mono, resampled to 22,050 Hz (soxr), 16-bit.
2. **Text**: English transcripts normalised by the English frontend rules for the MFA dictionary; Telugu transcripts by the Telugu normaliser; one pronunciation dictionary per language generated from the project's own frontends, so training and inference use the same phone sets.
3. **Forced alignment** (MFA 3.4.3): English 3,821 / 3,821 utterances aligned; Telugu 2,586 / 2,614 measured. The English aligner is MFA's pretrained model and saw none of our data. The Telugu aligner had to be trained here, and it was trained on all 6 selected hours, validation included, because the aligner itself produces the validation duration targets. This is unsupervised alignment, as in the reference recipe (which aligns its whole corpus); the TTS model never trains on validation audio or text. It does mean the Telugu validation durations come from an aligner that saw those clips, so validation duration errors may be slightly optimistic estimated.
4. **Features** (reference preprocessor with fixes M5–M7): leading/trailing silence trimmed using the alignment; 80-band log-mel (FFT 1024, hop 256 = 11.6 ms, 0–8 kHz, matching the HiFi-GAN vocoder); F0 with WORLD DIO + StoneMask, unvoiced frames linearly interpolated, averaged per phone; energy = L2 norm of each STFT frame, averaged per phone; pitch and energy z-normalised with the training-set mean/std; per-phone durations in frames from the TextGrids.

3. Training configuration

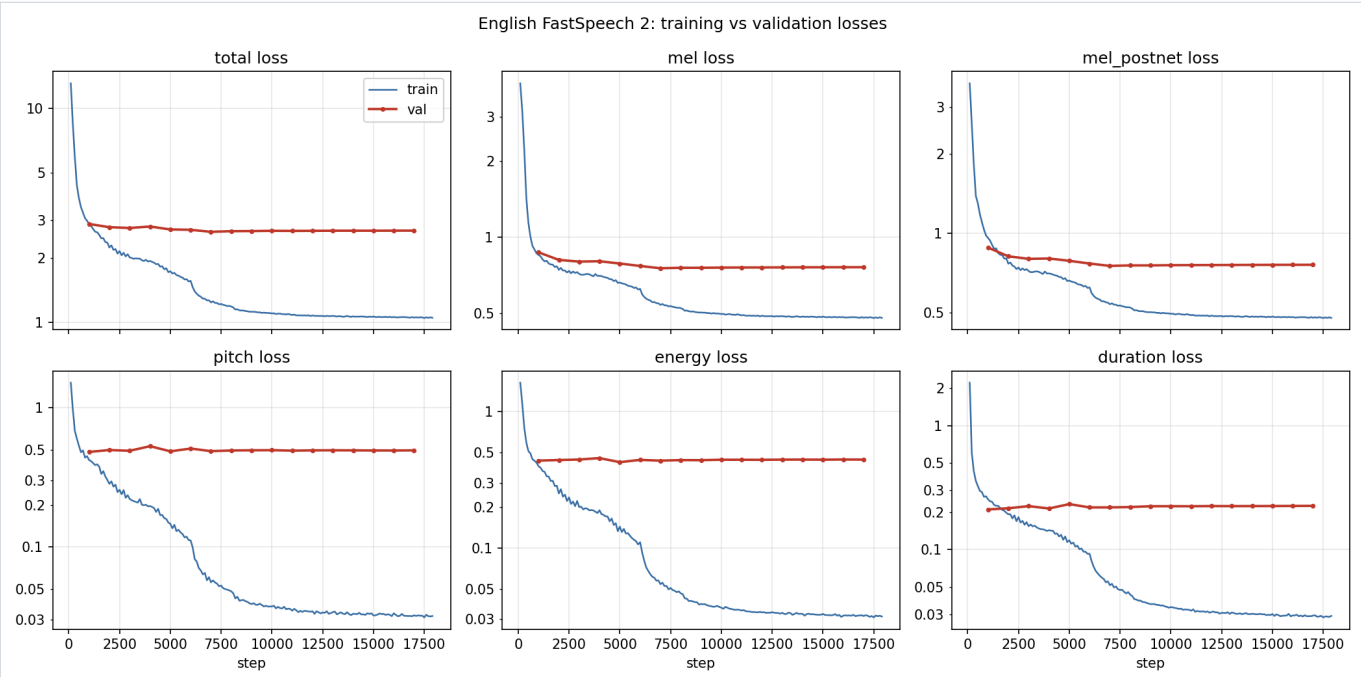
	English	Telugu
Model	reference architecture (§1.1); 35.44 M (en) / 35.34 M (te) parameters, the difference being the symbol tables (360 vs 65 symbols)	
Optimiser	Adam, β = (0.9, 0.98), ε = 10^{-9}; Noam warm-up 4,000 steps; ×0.3 decay at 1/3, 4/9 and 5/9 of the planned steps (the reference's 300k/400k/500k of 900k); gradient clipping 1.0; batch 16 (length-sorted groups of 4 batches); fp32	
Planned steps / LR decays	20,000 / 6k, 8k, 11k	56,000 / 18k, 24k, 31k
How the plan was set	300-step probe: 1.45 steps/s on 1 × T4; 210-min budget	300-step probe: 2.08 steps/s on 2 × T4; 480-min budget
Completed	17,949 steps in 211 min (stopped by budget)	55,524 steps in 480 min (stopped by budget)
Validation	every 1,000 steps on the full validation set: 6 losses + duration MAE and total-length error (`val_diagnostics.csv`)	

The English budget was cut from 8 h to 3.5 h for the rerun because the Telugu validation losses had stopped improving after ~8–10k steps (§4); this also saved GPU quota and shortened the exposure to another session failure. Exact configs: `English/config.yaml`, `Telugu/config.yaml`, and as run: `Results/<Lang>/train_config_used.yaml`.

4. Loss curves: convergence, stability and over-fitting



Telugu: training (blue, mean of each 100 steps) and validation (red, every 1,000 steps), log scale. Separate `training_loss.png` and `validation_loss.png` are in `Results/Telugu/`.



English, same layout (`Results/English/`).

Validation loss	English best (step)	English final	Telugu best (step)	Telugu final
total	2.651 (7k)	2.686	1.970 (16k)	1.974
mel after PostNet	0.753	0.760	0.589	0.595
pitch (z-units^2)	0.483	0.495	0.537	0.538
energy	0.427	0.445	0.192	0.192
log-duration	0.211	0.226	0.052	0.054

final *training* total	1.049	0.707

Convergence. Training losses fall fast for ~2k steps (warm-up), then steadily; visible steps in the training curves coincide with the learning-rate decays (Telugu at 18k, English at 6k), after which training loss flattens. Validation losses reach within 2 % of their final value by step 7k (Telugu) and 5k (English) and are flat afterwards measured.

Stability. No spikes, no divergence, no non-finite losses in either run (the trainer stops on any non-finite value; it never triggered).

Over-fitting. The gap between training and validation grows throughout: by the end the training pitch and energy losses are 30–40× lower than validation for Telugu, ~15× for English. This is memorisation of 5 h of data, mainly by the variance predictors. For Telugu it is benign: validation never got worse (final within 0.2 % of best). For English there is **mild over-fitting**: validation pitch, energy and duration losses drift upward by 3–7 % after 7k steps, and the total by 1.3 %. Early stopping on validation loss (here ~7–10k steps) would give a slightly better English model and save time; the released checkpoint is the last one estimated.

English vs Telugu level. English validation losses are higher in every term except pitch, consistent with the harder data: 110 speakers (≈ 2.7 min each), accented, more varied speaking style and recording conditions, against two studio-quality Telugu speakers with ~2.5 h each.

5. Generated speech, per test sentence

Audio: `Results/<Lang>/test_generated_audios/`. Voices: English `svarah_8a8e2655` (chosen by recording clarity, §5.0); Telugu `te_female` (lower HiFi-GAN copy-synthesis error than `te_male`: mel L1 0.378 vs 0.394 on 20 validation clips each) measured. To assess pronunciation without a listening panel, each wav was transcribed with Whisper large-v3 and compared with the frontend's normalised text; CER is shown both raw and after Whisper's English text normaliser (numbers, spellings), which removes pure formatting differences ("10:45" vs "ten forty five").

5.0 Choice of the English test voice, and where quality is lost measured

The first English outputs (voice `svarah_71f88d2a`, chosen because it has the most training audio) sounded muffled on listening. A Praat analysis (`tools/acoustic_analysis.py`) and a diagnostic job (`fastspeech2/diagnose_quality.py`, Kaggle `fs2-07-diagnose`, results in `Evaluation/Diagnosis/`) located the cause. For 24–30 validation clips of each test voice it compares the natural recording, copy-synthesis (ground-truth mel → HiFi-GAN), FastSpeech 2 with ground-truth duration/pitch/energy (oracle) and free-running FastSpeech 2, band by band (long-term spectrum, dB relative to the natural recording):

Band (Hz)	0–500	500–1k	1k–2k	2k–4k	4k–6k	6k–8k
English, copy − natural	−0.1	+0.7	+0.5	+0.8	+1.0	+1.1
English, free-running − natural	−0.3	+3.1	0.0	+3.1	+3.9	+1.9
Telugu, copy − natural	0.0	+0.4	−0.7	+0.1	+0.2	0.0
Telugu, free-running − natural	+0.3	−0.3	−2.4	−1.9	−1.1	−0.2

Neither the vocoder nor the acoustic model removes high-frequency energy: both reproduce the recording to within ~1 dB (copy) and ~4 dB (free-running) in every band, and HNR, jitter and shimmer of all four versions are within 1.5 dB / 0.4 % / 1.2 % of each other. The muffled sound was **the voice's own recordings**: ranking all 110 training speakers by the clarity of their recordings (1–4 kHz band share + HNR/2, three training clips each, `clarity_en.json`) put `svarah_71f88d2a` 108th (1–4 kHz band −23 dB). Svarah is crowd-sourced; recording conditions vary by speaker. The test sentences were rendered in the four clearest voices; `svarah_8a8e2655` (rank 2, 1–4 kHz band −9.7 dB) had the lowest jitter and shimmer of the four and is used for all English outputs. The checkpoint is unchanged; only the speaker embedding chosen at inference differs. Effect on the test sentences (means over E1–E4): spectral centre of gravity 300 → 620 Hz, jitter 1.2 → 0.5 %, shimmer 11.1 → 9.0 %; Whisper CER (normalised) on E1 / E3 / E4 from 15 / 17 / 40 % to 0 / 0 / 3 %. The selection rule was changed after listening to the first outputs; it is an objective rule on training recordings, and the earlier outputs remain in the repository history (commit `8f5ae5f`).

5.1 English

ID	Duration	Frontend decisions	Whisper transcript of our audio	CER raw / norm.
E1	6.32 s	*antidisestablishmentarianism*: not in the lexicon → g2p_en `AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M` (plausible, primary stress on -tar-)	exact ("anti-disestablishmentarianism")	0 / 0 %
E2a	4.24 s	*conduct* tagged noun (NN) → `K AA1 N D AH0 K T` (stress on 1st syllable)	exact	0 / 0 %
E2b	4.55 s	*conduct* tagged verb (VB) → `K AA0 N D AH1 K T` (stress on 2nd syllable)	"The researchers will conduct an experimental evaluation of the proposed method."	10.9 / 10.9 %
E3	2.88 s	"rhyth m" joined for pronunciation → *rhythm* `R IH1 DH AH0 M`; text unchanged	exact	0 / 0 %

E4	7.19 s	"ten forty five A M on twelve November twenty twenty six"; AM as letter names, durations ×1.6; *andaroo* (Telugu అందరూ "everyone") → `AE0 N D AA1 R UW0`	"The meeting will start at 10.45 am on 12 November 2026, so and are should arrive on time."	37.6 / 2.9 %

Observations. Four of the five sentences are transcribed exactly after normalisation (E4's raw CER is digit formatting: "10.45", "12", "2026"). The long rare word in E1 is recognised as the intended word. The homograph pair is pronounced as intended (E2a exact; in E2b "experiment to evaluate" is heard as "experimental evaluation of", a word-boundary slip in a fast unstressed stretch, not a stress error on *conduct*). E3 is exact, including the joined *rhythm*. E4: numbers, date and *AM* are correct; *AM* was heard as "pm" while its letters were predicted in 244 ms, and is heard correctly after the letter-name duration rule (453 ms). *andaroo*, a Telugu word in English text, is heard as "and are": its phones are right, but the ASR has no such word. Remaining audible weaknesses, from the per-word Praat analysis: unstressed function words are short and breathy (*the* 93–116 ms, HNR 11–15 dB against a sentence median of 16–17 dB), "twenty twenty" is fast (20–23 phones/s against ~11) and phrase-final words are slightly breathy (*arrive*, *time*).

Before/after the frontend fix. The first full English run (same checkpoint) read E3 as `R AY1 DH` + the letter name `EH1 M` ("rythe em") and ended *andaroo* in `OW`. Both were caught by inspecting `frontend_trace.json`, fixed by general rules (not word-specific exceptions), covered by tests, and the English test outputs were regenerated with the unchanged checkpoint (Kaggle job `fs2-05-reinfer-en`).

5.2 Telugu

ID	Duration	Frontend decisions	Whisper transcript of our audio	CER
T1	5.27 s	all words by rule; `ny` before `c` in విశ్లేషించారు	Neuro-Pharmacological పరిసోధనలో నూతన ఉషధాల ప్రభావాన్ని విస్లేశించారు.	38.7 %
T2a	2.46 s	పట్టు → `p a tt tt u`	తాడు పయతని పట్టు చాలా బలంగా ఉంది.	7.1 %
T2b	2.69 s	పట్టు → `p a tt tt u` (same as T2a)	ఆమేపట్టు చిరను ధరించి వేడుకకు వచ్చింది.	5.9 %
T3	2.14 s	the split అడ విలో kept as two words; no pause between them	పులి అడవిలో నిస్సబ్ధంగా నడిచింది.	6.9 %
T4	9.13 s	2026 → రెండు వేల ఇరవై ఆరు; 12వ → పన్నెండవ; 10:45 → పది నలబై ఐదు	2026 నవం బర్పన్ నండవతేది ఉదయం 10.45 గంటలకు ప్రారంబమోతుంది సమయానికి సిద్ధంగా ఉంచండి	38.8 %

Observations. All five sentences are synthesised without errors in the frontend, with every number expanded to Telugu words. Against Whisper's Telugu recognition the clearest sentence is T2b (CER 5.9 %) and the hardest T4 (CER 38.8 %); the mean over the five is 19.5 %. Transcripts should be read with Whisper's weak Telugu ability in mind (below). What the transcripts show, sentence by sentence: **T1** — Whisper recognises the long English loanword and writes it in Latin script (*Neuro-Pharmacological*), so most of T1's CER is a script mismatch, not a mispronunciation; the remaining errors are శ → స in పరిశోధనలో and విశ్లేషించారు and ఔ → ఉ in ఔషధాల. **T2a/T2b** — పట్టు is transcribed correctly in both senses (the two are written and pronounced identically; only the pitch differs, §6.2); the errors are word-boundary merges (పయతని, ఆమేపట్టు) and vowel length (ఆమె → ఆమే, చీర → చిర). **T3** — the split అడ విలో is heard as the single word అడవిలో, i.e. the model joins it as a listener would; the only error is again శ → స (నిస్సబ్ధంగా). **T4** — Whisper writes the expanded numbers back as digits (2026, 10.45), which the CER counts as errors, drops the opening ఈ సమావేశం, and merges నవంబర్ పన్నెండవ తేదీ across word boundaries (the syllables are all present); ప్రారంభ → ప్రారంబ is a lost aspiration. The one systematic pattern is శ (frontend phone `sh`) heard as స (`s`) in all four words that contain it. The frontend keeps the two phones distinct, so the merger is in the acoustic model or in Whisper; with only these transcripts the two cannot be separated estimated.

Whisper's Telugu recognition is far weaker than its English recognition (its copy-synthesis ceiling on Telugu validation speech is 33.1 % CER, against 11.6 % for English), so Telugu CERs are an upper bound on pronunciation errors, not a direct measure of them.

6. Variance-adaptor plots, per test sentence

For each sentence, three plots (`Results/<Lang>/variance_adaptor_plots/`): **duration** — predicted frames per phone (bars; right axis in ms); **pitch** — predicted phone-level F0 in Hz (red steps over each phone's predicted time span) with the F0 that WORLD extracts from the generated wav (grey dots; unvoiced frames left as gaps), showing whether the vocoder output follows the prediction; **energy** — predicted phone-level energy (STFT-frame L2 norm, arbitrary units). There is no ground truth for test sentences. The numbers behind each plot are in `variance_raw/<ID>.json`.

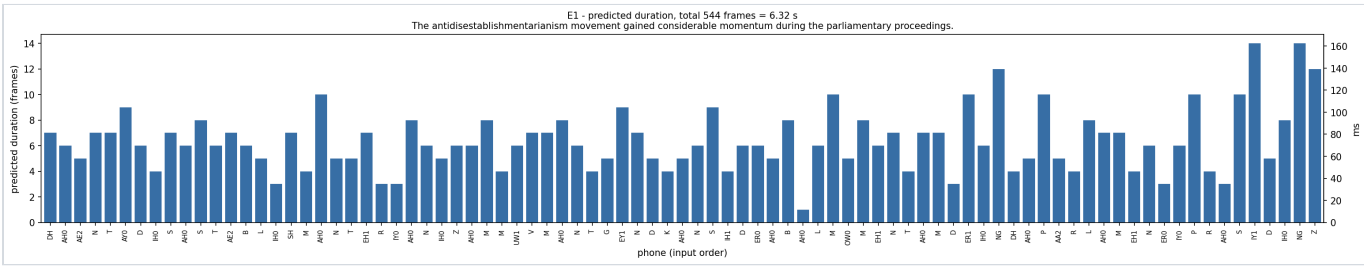
Patterns common to all ten sentences measured

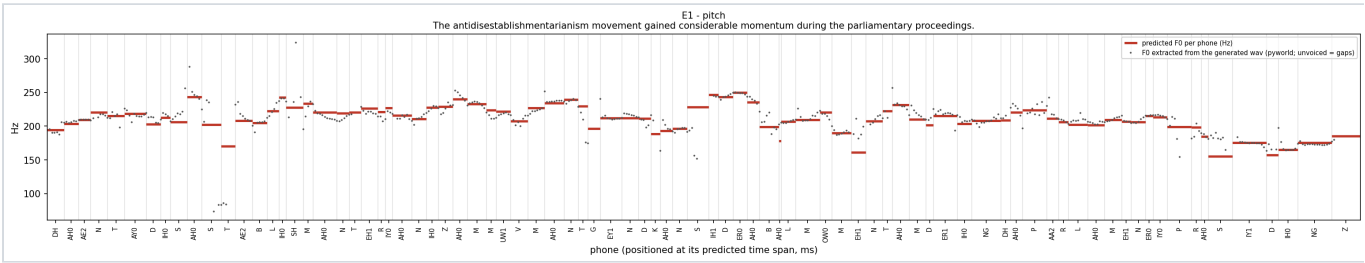
- **Declination**: pitch is highest on an early content word and lowest on the last word in every sentence (English first-third mean 214–235 Hz → last third 192–210 Hz; Telugu 217–246 → 178–197 Hz).
- **Phrase-final lengthening** in 9/10 sentences: the last vowel is 1.3–2.9× the sentence's mean vowel duration (E2a is the exception, 0.7×).
- **Segmental energy pattern**: vowels and nasals high, stops/fricatives and pauses low — e.g. E4 /S/ ≈ 8, vowels 80–150 a.u. — as expected for an STFT-norm energy.
- **Speaking rate** 11.7–13.5 phones/s (English), 12.5–14.5 phones/s (Telugu).
- The F0 measured on the generated audio tracks the predicted phone-level F0 closely within voiced phones, so the decoder and vocoder realise the predicted pitch; deviations appear at phone boundaries and on short phones, as expected from phone-level (step-wise) targets.

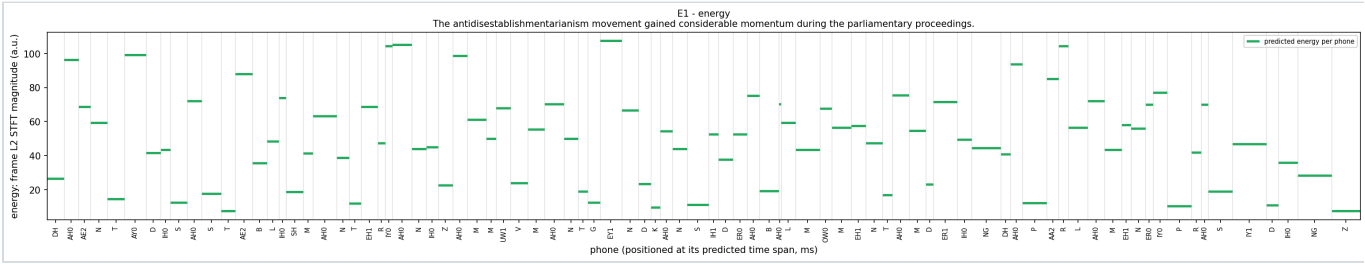
6.1 English

E1 — The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

- *antidisestablishmentarianism* takes 1,962 ms, 31 % of the 6.32 s sentence, with uniformly short phones and no pause before or after it.
- Pitch has several similar peaks (*considerable* 250 Hz, *antidisestablishmentarianism* 243, *movement* 239, *momentum* 232 Hz) and falls to its minimum on the last word *proceedings* (199 Hz): a flat, list-like contour with final declination (first third 216 → last third 200 Hz).
- *proceedings* is the longest word after the long word (929 ms) and its final vowel is lengthened 1.3×, a phrase-final cue. No pause is predicted, because the sentence has no internal punctuation.

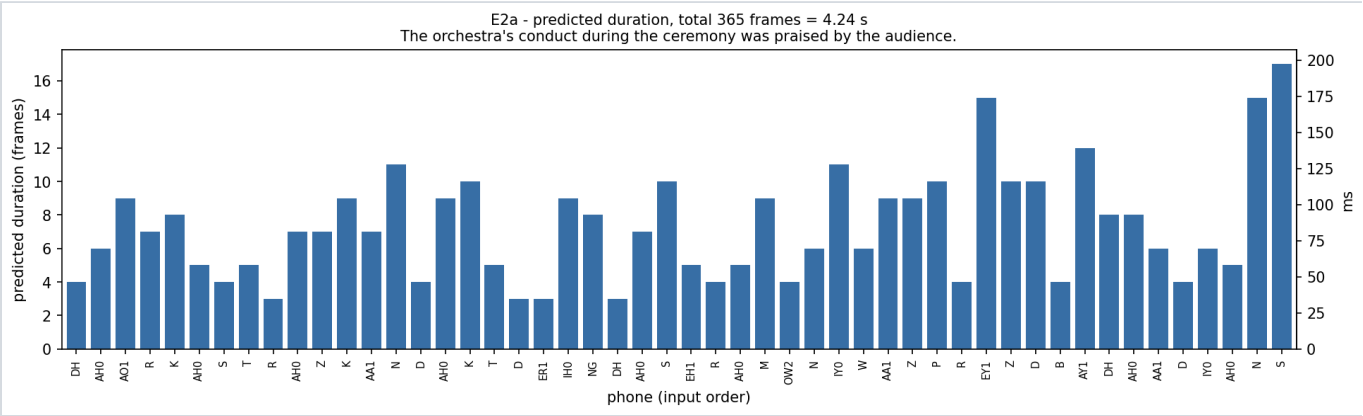


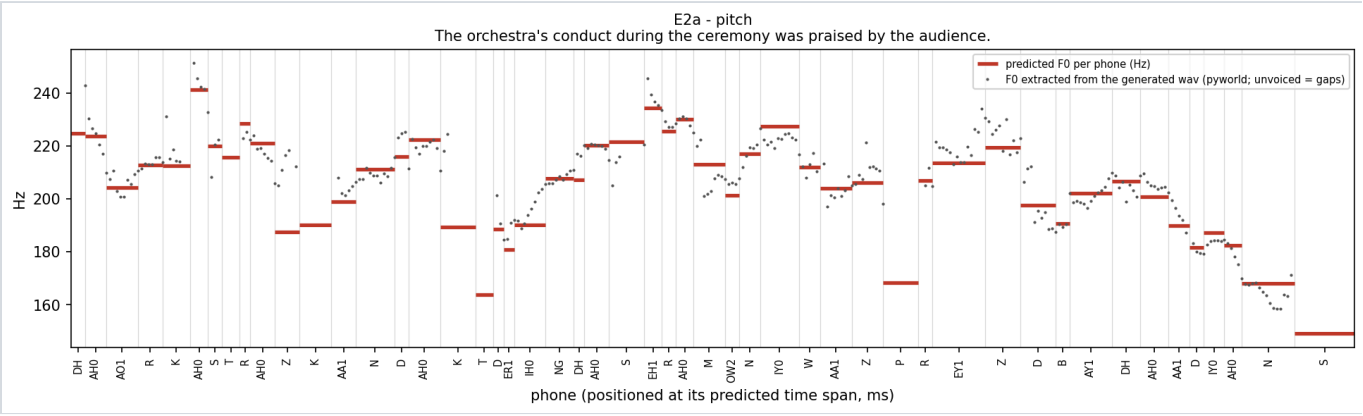


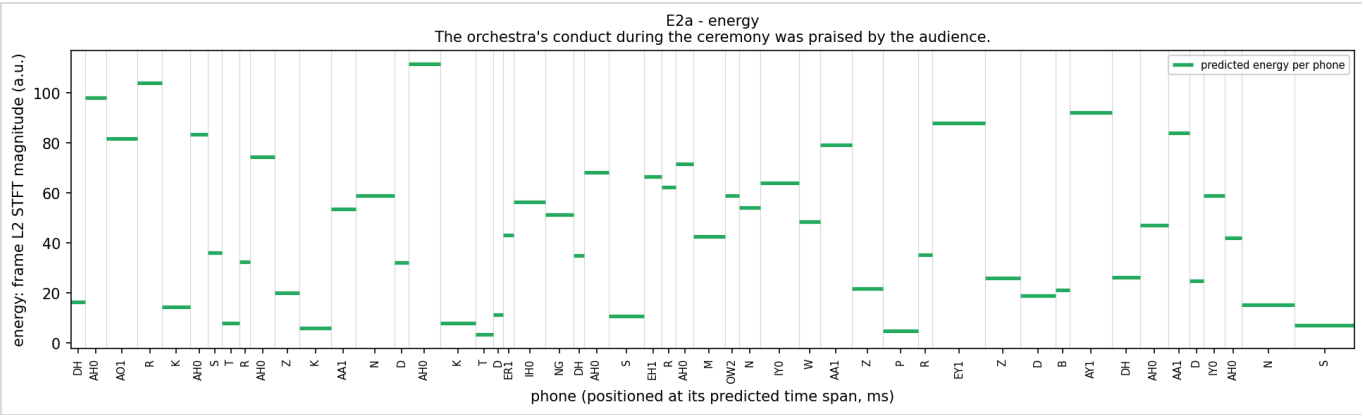


E2a — The orchestra's conduct during the ceremony was praised by the audience.

- *conduct* (noun, `K AA1 N D AH0 K T`): first syllable 313 ms vs second 325 ms (ratio 0.96). That is relatively more weight on the first syllable than the verb in E2b gets (0.82), but not an absolute first-syllable dominance. Pitch rises across the word (199 → 222 Hz on its vowels), so in this voice the noun stress is cued weakly, by relative duration only.
- Main accent on *orchestra's* (241 Hz) and *ceremony* (234 Hz), then a fall to 190 Hz on *audience*.
- The final vowel is short (0.7× the mean) because *audience* ends in /s/; the word as a whole is long (615 ms). *conduct* as a noun is 639 ms, the longest word of the sentence.

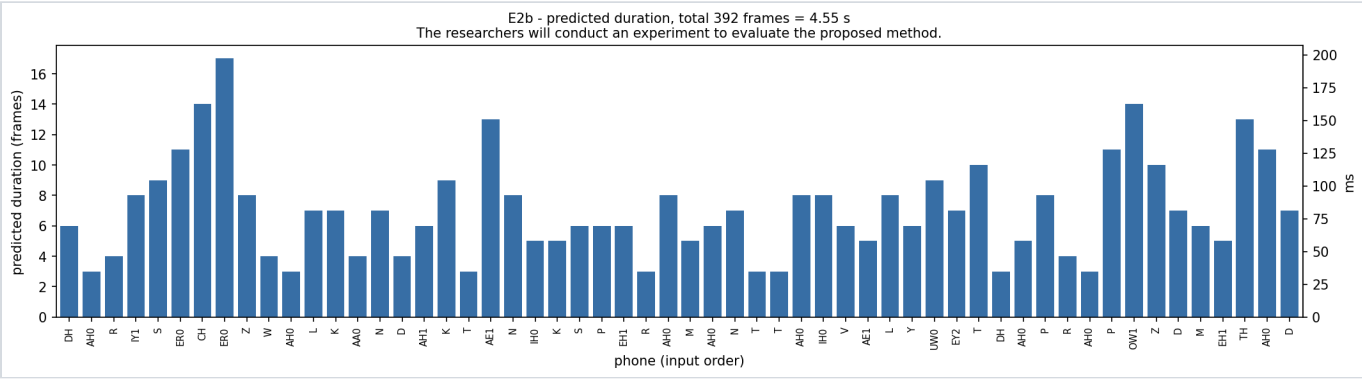


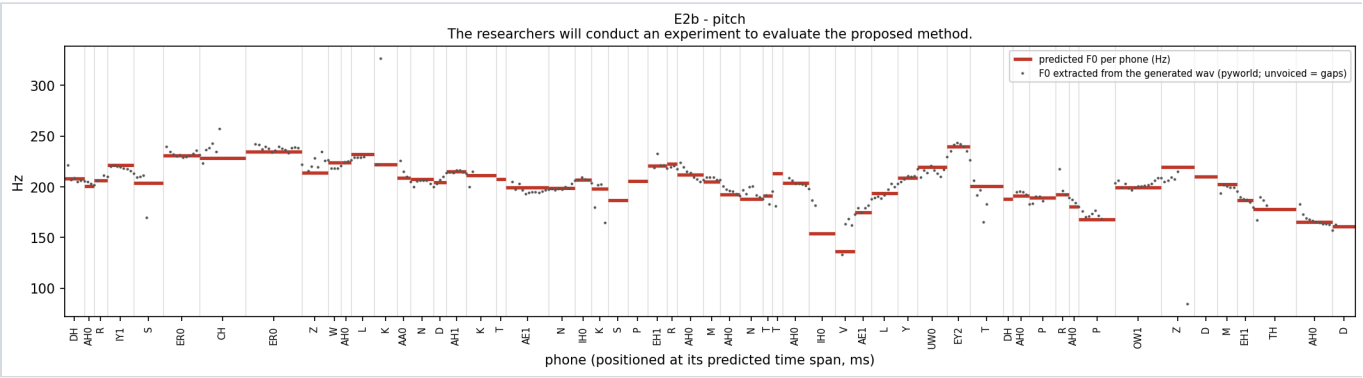


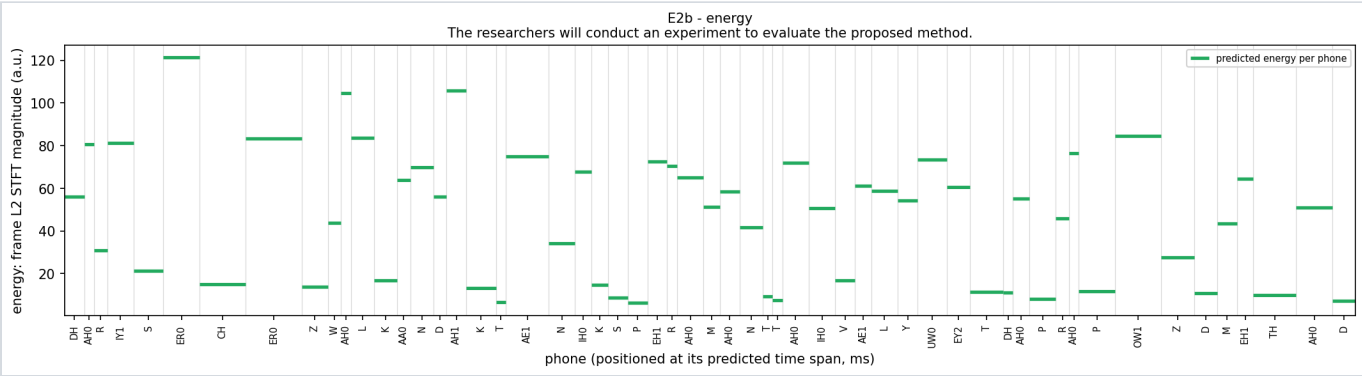


E2b — The researchers will conduct an experiment to evaluate the proposed method.

- *conduct* (verb, `K AA0 N D AH1 K T`): first syllable 209 ms vs second 255 ms (ratio 0.82, shifted towards the second syllable relative to E2a's 0.96), with the higher pitch on the stressed second vowel (AH1 215 Hz vs AA0 209 Hz).
- The word is 464 ms against 639 ms as a noun: the verb is less prominent in its phrase than the subject noun of E2a. The homograph contrast is carried mainly by duration.
- Peaks on *evaluate* (240 Hz) and *researchers* (235 Hz); declination to *method* (202 Hz) with 1.5× final lengthening. *to evaluate* is fast (128 ms for *to*), where the ASR heard "experimental evaluation".

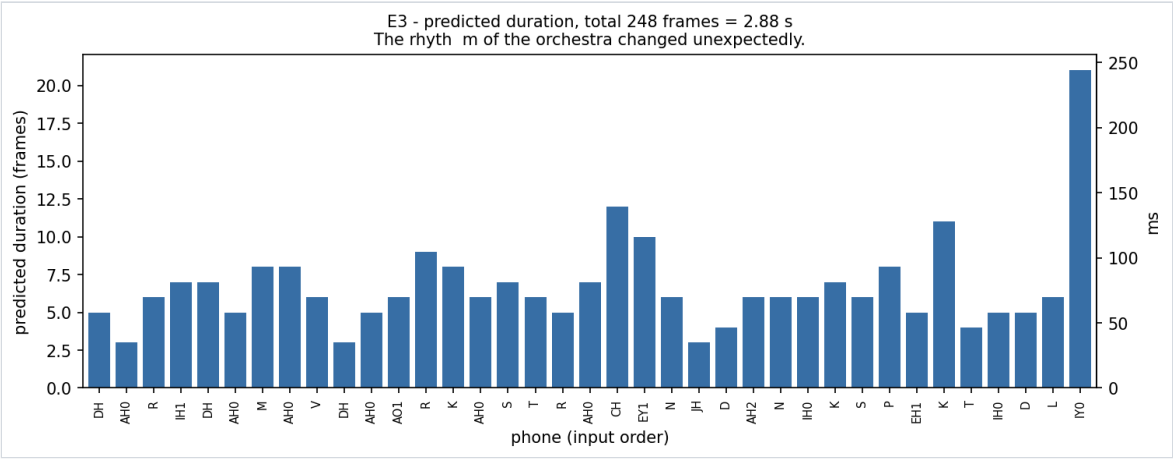


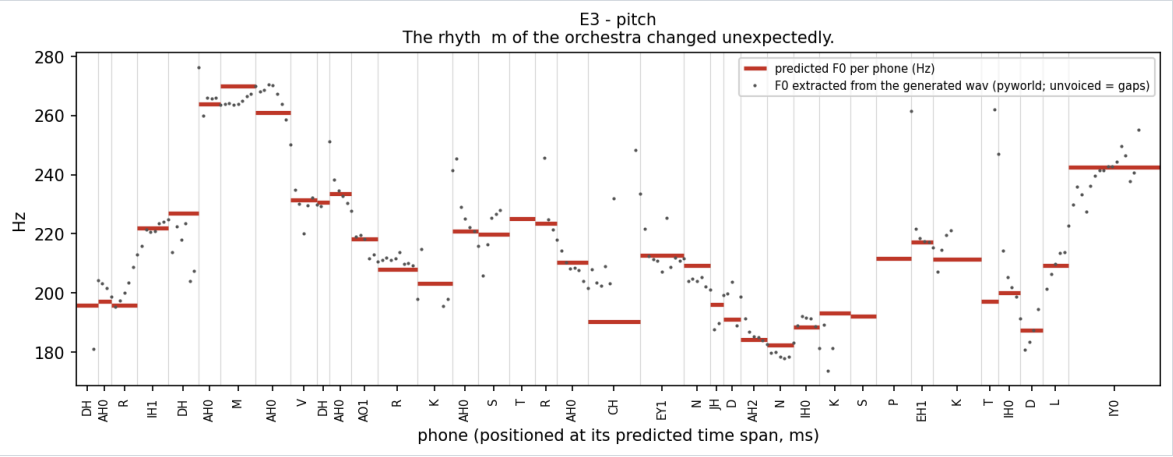


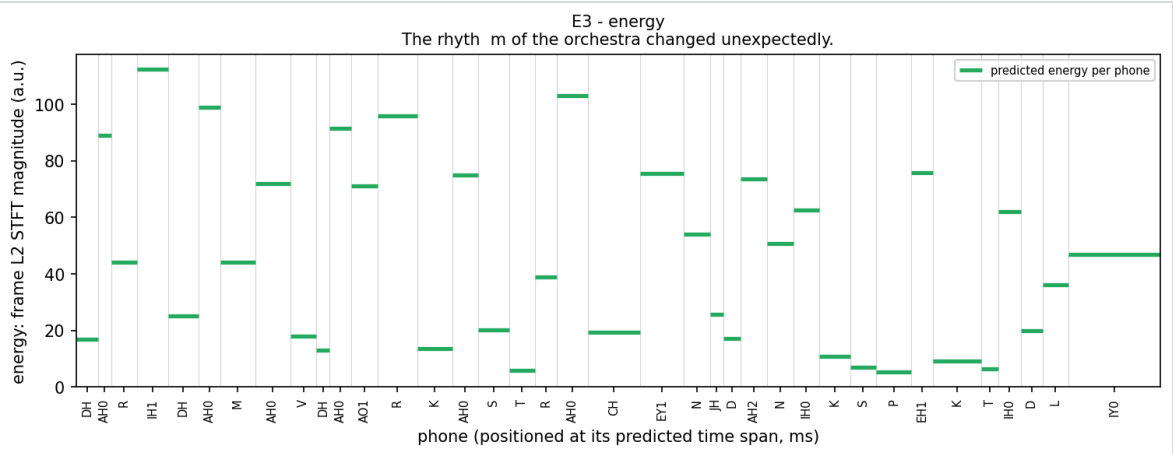


E3 — The rhyth m of the orchestra changed unexpectedly.

- The split word is pronounced as *rhythm* (`R IH1 DH AH0 M`, 383 ms) with the sentence's highest pitch (270 Hz, and 261 Hz on *of*): a strong initial accent.
- *changed* is low (213 Hz); *unexpectedly* is very long (1,115 ms), with a final vowel of 244 ms (2.9× the mean vowel), the strongest final lengthening of the ten sentences.
- No pause is predicted between *rhythm* and *of*; the double space in the text has no effect after normalisation.

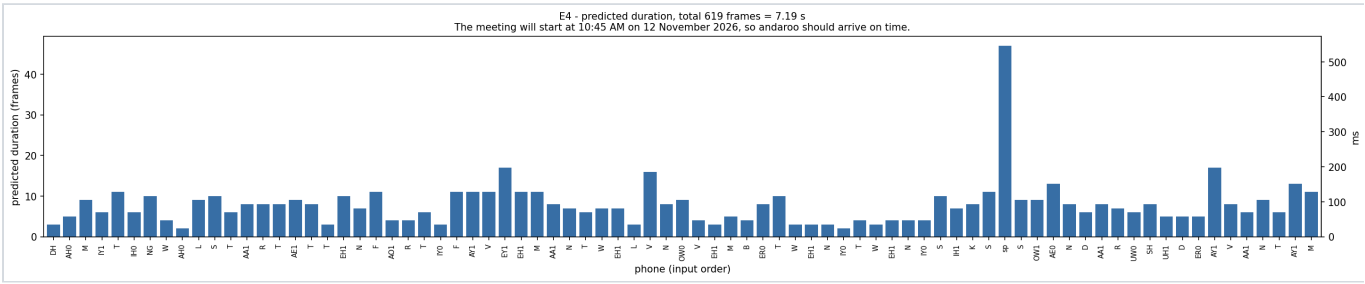


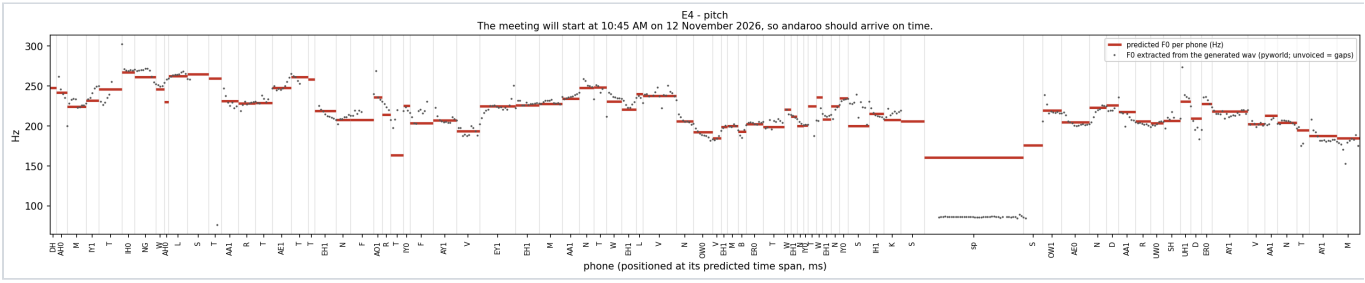


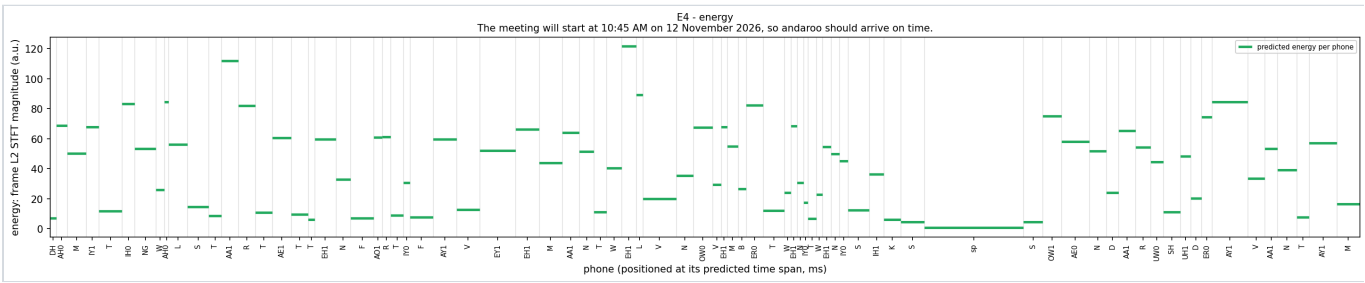


E4 — The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

- Numbers are read as words with growing durations (*ten* 232, *forty* 325, *five* 383 ms; *twelve* 453, *November* 476, *six* 418 ms): list-like, each number group ends long, while *twenty twenty* is fast (244 + 221 ms). *AM* takes 453 ms with the letter-name duration rule (244 ms without it).
- The comma yields a **546 ms pause** (7.6 % of the sentence; 824 ms with the former voice, so the speaker embedding changes pause length). After it there is no pitch reset (*six* 215 → *so* 219 Hz), and the contour declines to the sentence minimum on *time* (194 Hz); the highest pitch is on *meeting* (267 Hz).
- *andaroo* (`AE0 N D AA1 R UW0`) gets 557 ms with stress on the second syllable; final lengthening 1.8× on *time*.



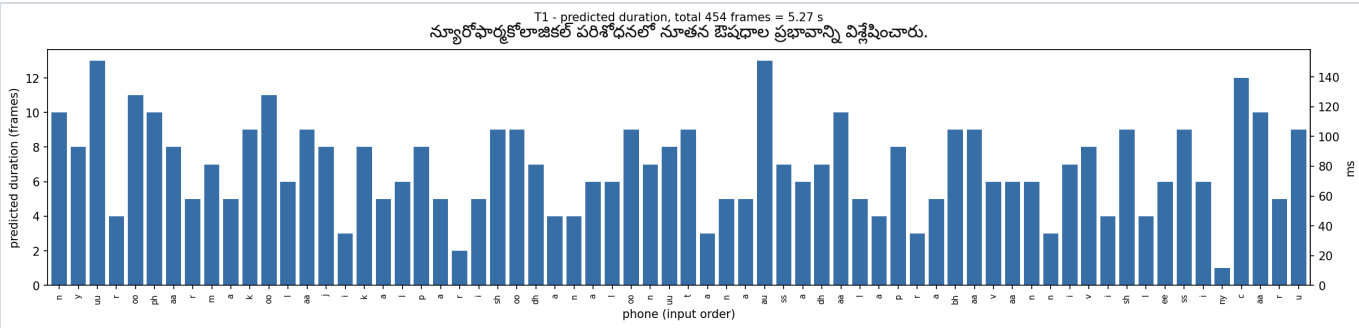


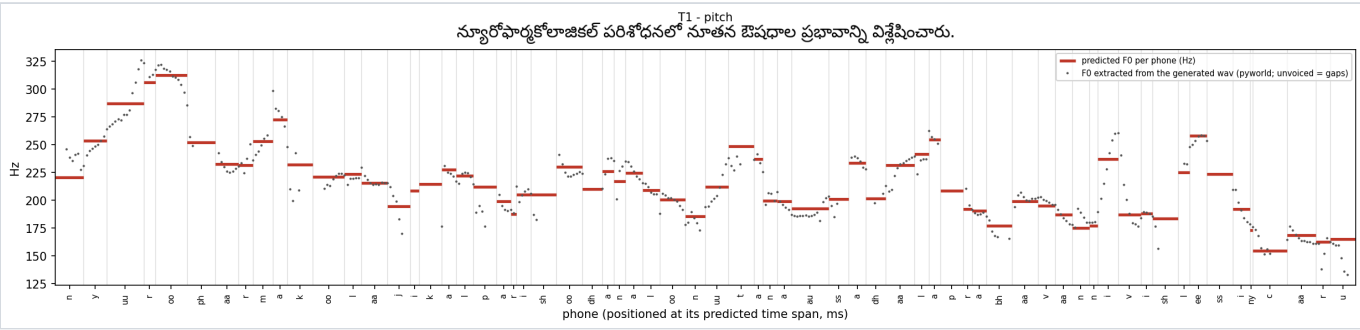


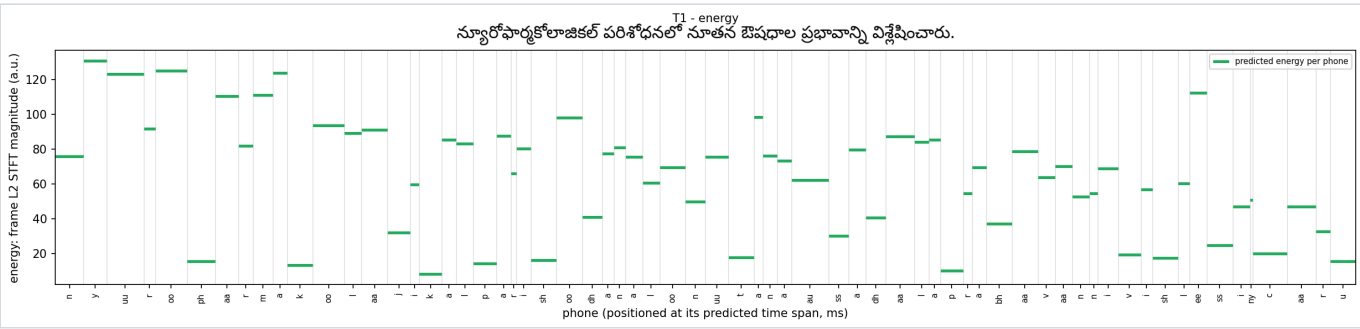
6.2 Telugu

T1 — న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- న్యూరోఫార్మకోలాజికల్ lasts 1,695 ms (32 % of 5.27 s) and carries the sentence's highest pitch (313 Hz): the long loanword is the focus of the sentence-initial phrase.
- Pitch drops to 230 Hz on పరిశోధనలో and stays in a narrow 237–258 Hz band for the rest: a typical Telugu (verb-final) declarative pattern with the main prominence early.
- The final verb విశ్లేషించారు is long (964 ms; cluster `sh 1`, nasal `ny` before `c`) with 1.3× final lengthening; no internal pauses (no internal punctuation).

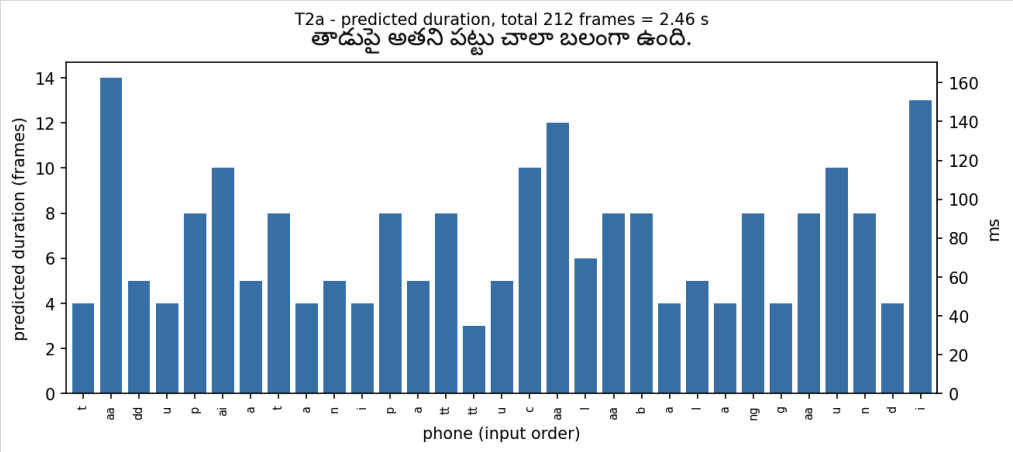


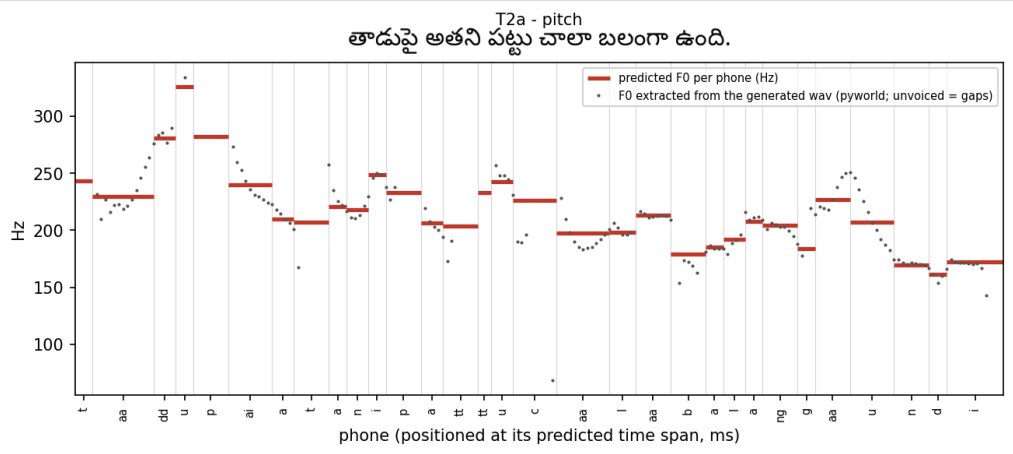


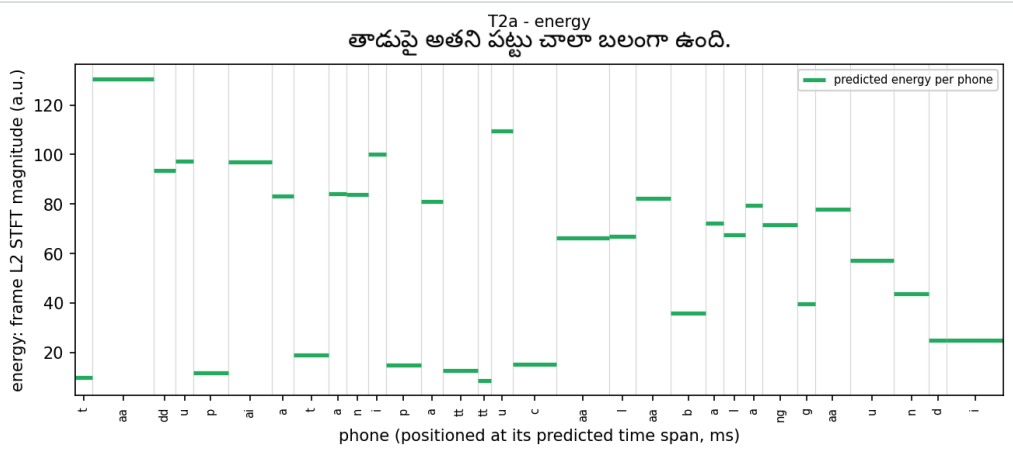


T2a — తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

- Peak pitch on the first word తాడుపై (326 Hz), then a steady fall to ఉంది (207 Hz).
- పట్టు (`p a tt tt u`, geminate retroflex) lasts 337 ms with 242 Hz — inside the falling line, not accented.
- Final vowel 151 ms, 1.8× the mean: strong phrase-final lengthening.

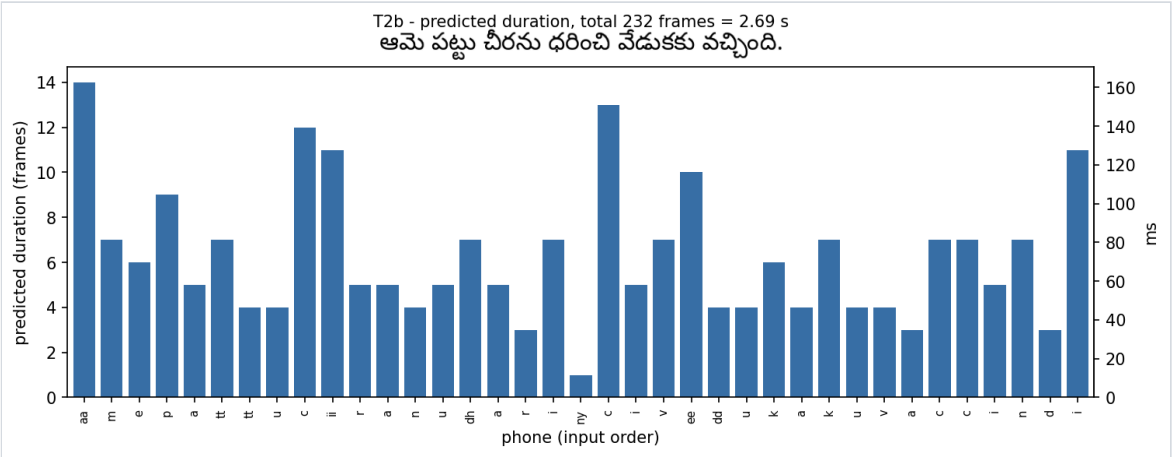


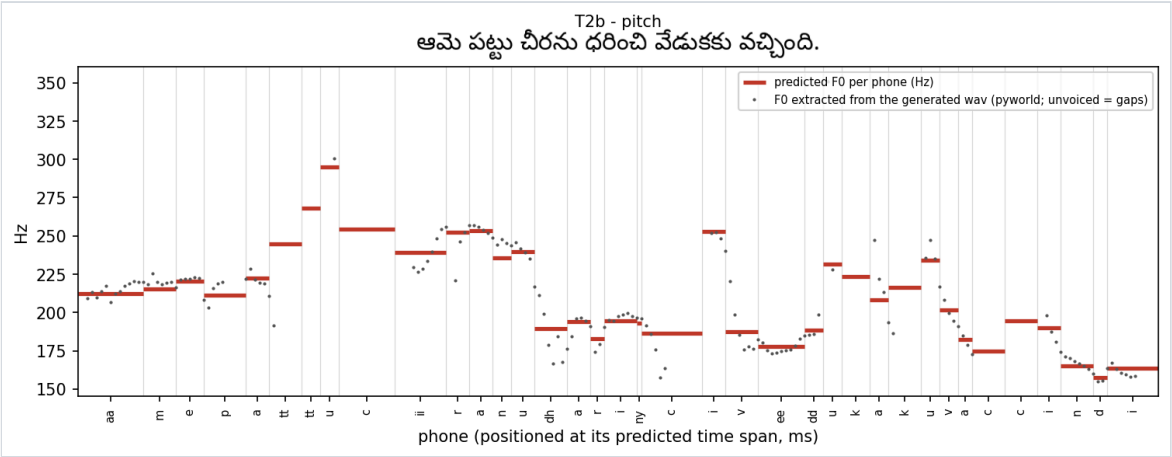


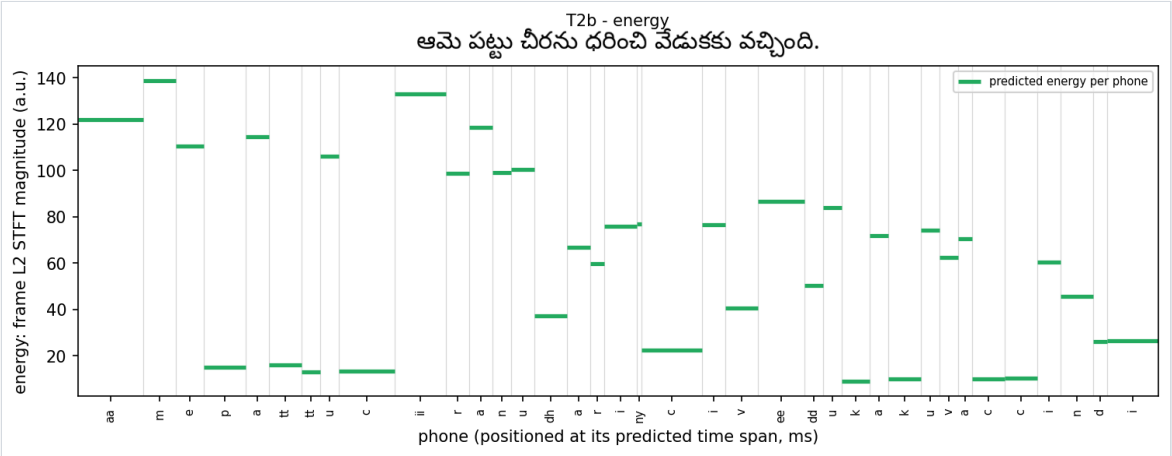


T2b — ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- పట్టు has exactly the same phones and the same duration (337 ms) as in T2a, as it should (the two words are homographs and homophones), but now carries the sentence's pitch peak (295 Hz vs 242 Hz in T2a): as the modifier of the compound పట్టుచీర (silk saree) it is accented, after a low start on ఆమె (220 Hz).
- The rest declines from 254 Hz (చీరను) to 202 Hz (వచ్చింది), with 1.7× final lengthening.
- The difference between T2a and T2b is therefore purely prosodic (position and focus), which the model reproduces.

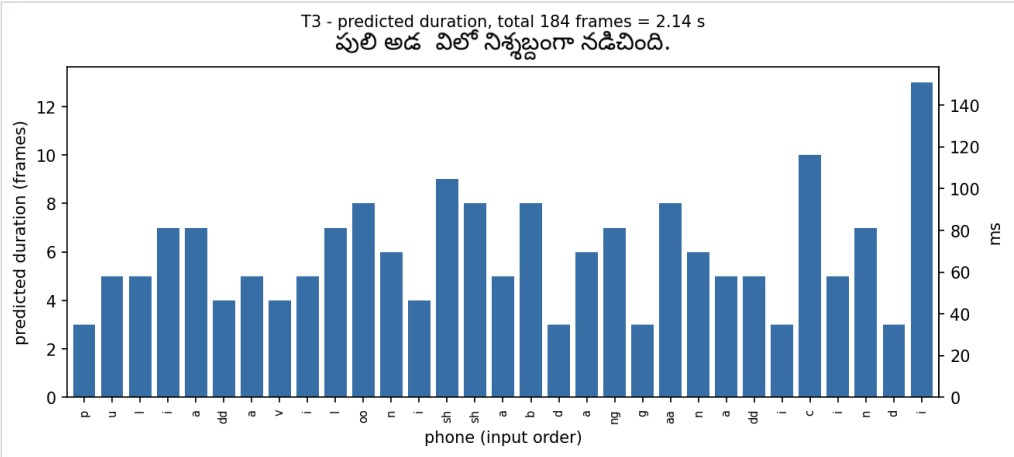


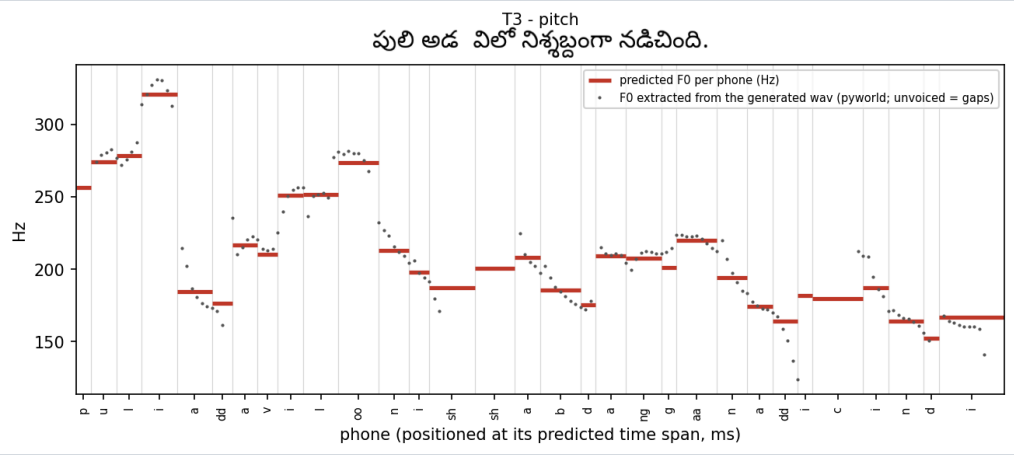


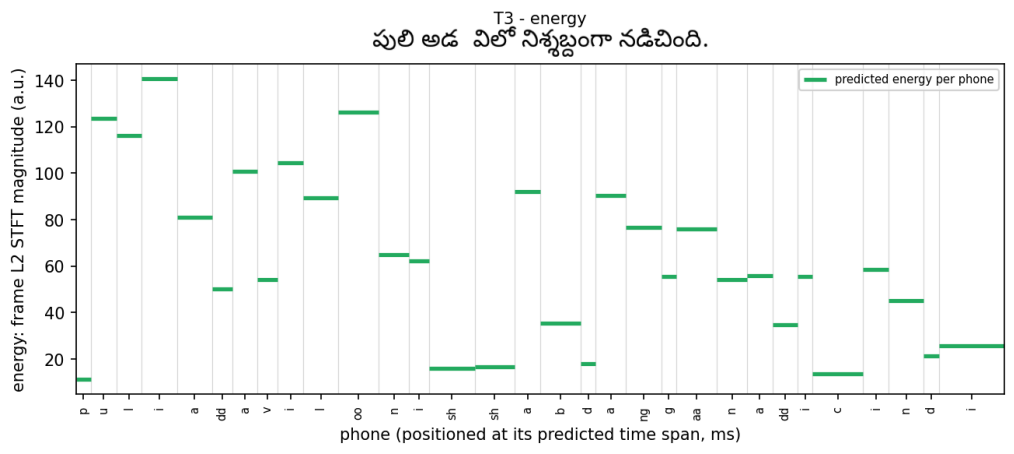


T3 — పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- Peak on పులి (321 Hz, the topic).
- The broken word అడ విలో (for అడవిలో) gets no pause (అడ 186 ms, విలో 279 ms, contiguous), but pitch rises from 217 Hz on అడ to 274 Hz on విలో: the model treats విలో as a new word onset, so the split is partly audible as a prosodic break.
- నిశ్శబ్దంగా is long (778 ms; geminate `sh sh` and the cluster `b d`); నడిచింది ends the sentence at the pitch minimum (194 Hz) with 2.1× final lengthening.

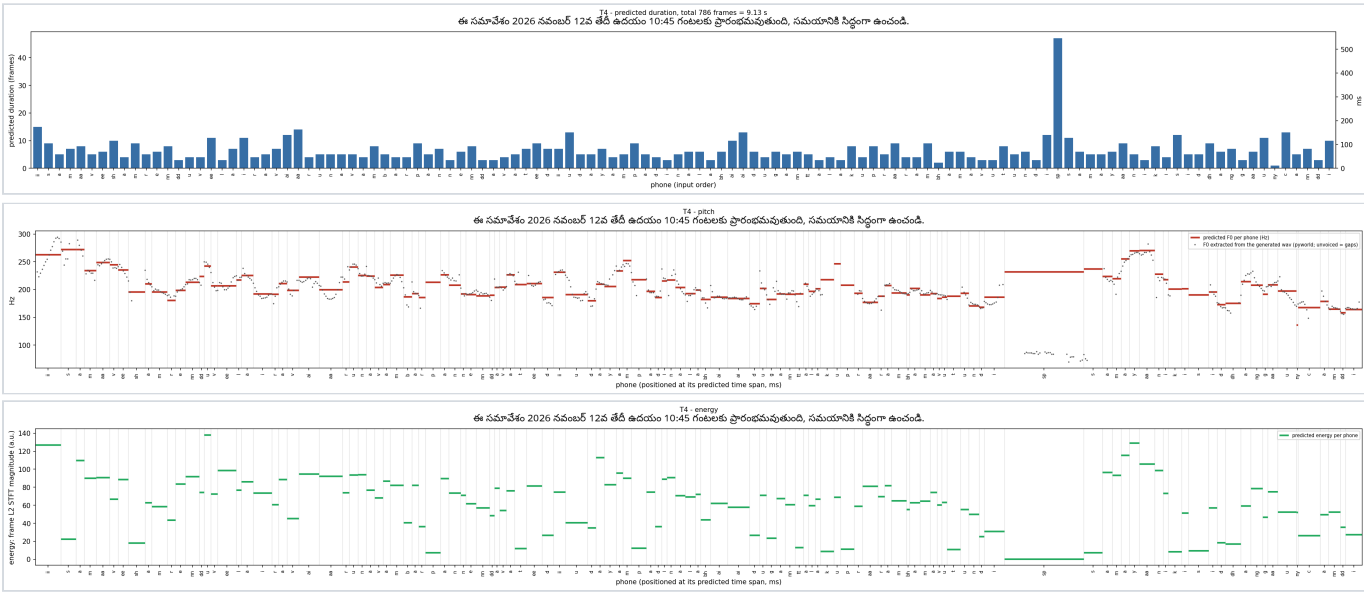






T4 — ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

- Longest sentence (9.13 s, 120 phones). Numbers are expanded to Telugu words: రెండు వేల ఇరవై ఆరు (302/290/453/267 ms), పన్నెండవ (615 ms), పది నలభై ఐదు (244/418/267 ms), read without hesitation pauses.
- The comma gives a 546 ms pause followed by a clear **pitch reset**: from ≈ 208 Hz on ప్రారంభమవుతుంది (the longest word, 1,126 ms) up to 270 Hz on సమయానికి — the second clause starts a new intonation phrase.
- Highest pitch on సమావేశం (272 Hz), lowest on the final ఉంచండి (198 Hz), with 1.5× final lengthening.



7. Objective evaluation on held-out data

Beyond the required outputs, both models were evaluated on **every validation utterance** (never trained on), each synthesised with its own speaker (`fastspeech2/evaluate_val.py`, Kaggle `fs2-04-eval`, `fs2-06-eval-fix`). Silence phones are excluded from duration/pitch/energy metrics.

Metric measured	English (742 utt.)	Telugu (446 utt.)
Duration MAE (frames, 11.6 ms) / Pearson r of log(1+d)	2.81 / 0.58	1.06 / 0.85
Total utterance length, relative error	11.7 %	4.5 %
Phone-level pitch RMSE / Pearson r	34.6 Hz / 0.35	29.2 Hz / 0.76
Phone-level energy RMSE / Pearson r	26.8 / 0.78	17.4 / 0.90
Mel-cepstral distortion*, ground-truth duration/pitch/energy (DTW)	36.6	26.1
Mel-cepstral distortion*, predicted duration/pitch/energy (DTW)	39.0	30.9
Whisper large-v3 CER / WER, our synthesis (150 utt.)	15.7 % / 21.5 %	35.8 % / 80.3 %
Same, copy-synthesis ceiling (ground-truth mel → same HiFi-GAN)	11.6 % / 13.8 %	33.1 % / 78.6 %

* Computed from the model's own 80-band log-mel (orthonormal DCT, c1–c24), averaged over DTW-aligned frames. This is not the WORLD/SPTK mel-cepstrum MCD reported in papers; it is comparable only within this table. CER/WER are corpus-level (total edits / total reference length) after lower-casing and removing punctuation.

Reading the table. (1) The variance predictors cost little spectral accuracy: replacing ground-truth duration, pitch and energy by predictions raises the distortion by 2.4 (en) and 4.8 (te) — most of the remaining error is in the decoder itself. (2) Telugu prosody is predicted far better than English (pitch r 0.76 vs 0.35; duration r 0.85 vs 0.58): with two speakers and read studio speech, phone identity and context determine prosody much more than in 110-speaker, partly spontaneous Svarah. (3) For English the intelligibility gap to the vocoder ceiling is 4.1 points of CER: the acoustic model loses some intelligibility, but most of the error budget is shared with copy-synthesis (accented speech, ASR limits).

7.1 Speed measured

Real-time factor (compute s / audio s), batch 1, warm	English	Telugu
GPU (Tesla T4): acoustic model / HiFi-GAN / total	0.0079 / 0.0193 / 0.027	0.0070 / 0.0185 / 0.025
CPU (4 threads, fp32): acoustic / HiFi-GAN / total	0.029 / 0.322 / 0.350	0.026 / 0.355 / 0.381

On a T4 one second of speech takes ~26 ms (≈ 37–40× faster than real time); the paper reports RTF 1.95×10^{-2} on a V100 with Parallel WaveGAN paper. On a CPU the acoustic model is still ~35× faster than real time; the vocoder dominates (~90 % of the time), so a lighter vocoder is the lever for CPU deployment. Timing excludes the text frontend; CUDA is synchronised before each reading; one untimed warm-up call.

7.2 Prosody controls measured

Control (factor)	English: length / median F0 / RMS	Telugu: length / median F0 / RMS
none	4.55 s / 205 Hz / 0.121	2.46 s / 209 Hz / 0.148
duration 0.8 / 1.25	3.37 s / 5.42 s	1.88 s / 3.02 s
pitch 0.8 / 1.2 (after M12)	169 Hz / 244 Hz	165 Hz / 252 Hz
pitch 0.8 / 1.2 (reference code)	140 Hz / 113 Hz — inverted (former voice, 126 Hz)	204 Hz / 209 Hz — almost no effect
energy 0.8 / 1.2 (after M12)	RMS 0.103 / 0.139	RMS 0.120 / 0.159

The duration control scales length close to the factor (0.74× and 1.19× for English; 0.76× and 1.23× for Telugu; rounding of per-phone frames explains the difference). Pitch control after M12 is near-proportional (×0.79/×1.20 for Telugu).

7.3 Stress sentences (beyond the assignment)

Sixteen extra sentences (`Evaluation/stress_sentences.json`, audio in `Evaluation/<Lang>/stress/`) probe cases the test set does not. All 16 went through the frontends without errors; the Whisper transcripts show what is intelligible:

English (CER after Whisper's English normaliser, against the original text)

ID	Case	Text	Whisper transcript	CER
SE01	time and numbers	The train leaves at 7:30 PM from platform 3.	The train leaves at 7.30 pm from Platform 3.	0.0 %
SE02	question intonation	Did you remember to bring the documents for tomorrow's interview?	There you remember to bring the documents for tomorrow's interview.	9.1 %
SE03	homograph (wind: noun / verb)	The wind was strong, so we had to wind the rope tightly.	The wind was strong, so we had to wind the rope tightly.	0.0 %
SE04	Indian names and places	Priya and Venkatesh travelled from Hyderabad to Chennai last weekend.	Priya and Venkatesh travelled from Hyderabad to Chennai last weekend.	0.0 %
SE05	rare technical words	The cryptographic protocol relies on elliptic curve signatures.	The cryptographic protocol relies on elliptic signatures.	9.1 %
SE06	abbreviations (known gap: no abbreviation expansion)	Dr. Rao presented the results at IIIT Hyderabad.	Dr. Prabhupada said that the result is saturated herbide.	55.8 %
SE07	long sentence	Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time, thanks to the new irrigation canals that were completed in early spring.	Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time thanks to the irrigation canals that were completed in early spring.	1.8 %
SE08	exclamation, very short	What a wonderful surprise!	What a wonderful surprise!	0.0 %

Telugu (CER against the normalised text)

ID	Case	Text	Whisper transcript	CER
ST01	question intonation	మీరు రేపు ఉదయం ఆఫీసుకు వస్తారా? Will you come to the office tomorrow morning?	మీరు రేపు ఉదైయమాఫి సుకువస్తారా.	15.4 %
ST02	numbers	ఈ పుస్తకం ధర 250 రూపాయలు. The price of this book is 250 rupees.	ఈ పుస్తకం ధర రేండు వందల యాభి రూపాయలు.	6.7 %
ST03	time	రైలు సాయంత్రం 6:30 కి బయలుదేరుతుంది. The train departs at 6:30 in the evening.	రేలు సాయంత్రమారు ముప్పైకి బయలు దేరుతుంది.	8.3 %
ST04	consonant clusters (conjuncts)	స్త్రీ శక్తి రాష్ట్ర అభివృద్ధికి ముఖ్యం. Women's strength is important for the state's development.	3. సక్తి రాష్ట్ర ఆభివరుద్ధి కి ముఖ్యమ్	37.1 %
ST05	English loanwords in Telugu script (with ZWNJ)	నా లాప్‌టాప్ బ్యాటరీ త్వరగా అయిపోతోంది. My laptop battery is draining quickly.	4 లాప్టాప్ బ్యాటరి త్వరగా ఆయి పోతోంది.	12.1 %

ST06	Latin-script word inside Telugu (code-mixing)	నేను నిన్న కొత్త iPhone కొన్నాను. I bought a new iPhone yesterday.	నేను నిన్న కొత్త ఆఈ ఫోన్ కొన్నాను.	21.4 %
ST07	long sentence	హైదరాబాద్ నగరంలో ప్రతి సంవత్సరం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాఠకులు, రచయితలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు. Thousands of readers, writers and students enthusiastically attend the book fair held every year in Hyderabad.	AIDARABADNAGARAN లో ప్రతి సంవచ్రం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాదకులు, రచైతలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు.	20.2 %
ST08	exclamation, very short	ఎంత అందమైన దృశ్యం! What a beautiful sight!	ఎంతా అందమైన ద్రుస్యం.	33.3 %

8. English vs Telugu; limitations, failures and improvements

8.1 Comparison

	English	Telugu
Data character	110 speakers, ~2.7 min each, Indian-accented, varied recording and style	2 speakers, ~2.5 h each, studio read speech
Text → phones	irregular spelling: lexicon + neural G2P; homographs, OOV words and normalisation are the main risks	near-phonemic script: deterministic rules cover every word; risks are normalisation and corrupted input text
Prosody prediction (validation)	weak: pitch r 0.35, duration r 0.58	good: pitch r 0.76, duration r 0.85
Over-fitting	mild (validation drifts up after 7k)	none visible (flat after ~10k)
Phrase structure	pause at the E4 comma (546 ms), no pitch reset after it (*six* 215 → *so* 219 Hz)	shorter pause at the T4 comma (546 ms) followed by a pitch reset (208 → 270 Hz)
Main failures on the test set	short, breathy function words; fast "twenty twenty" (E4); a word-boundary slip in E2b; the muffled first test voice (§5.0)	ఴ heard as ఌ in every word that has it; vowel-length and word-boundary errors; highest ASR error on T4 (CER 38.8 %, partly digits written by Whisper); see §5.2

8.2 Limitations

- **5 h of data** per language (the assignment's budget) against 24 h in the paper; the variance predictors memorise the training set (§4) and English prosody generalises poorly.
- **No listening test**: quality statements rely on ASR and acoustic measurements. Whisper is weak on Telugu, so Telugu intelligibility is under-measured.
- **Universal vocoder, not fine-tuned**: copy-synthesis costs 11.6 % CER on English validation speech, but its spectrum matches the recordings within ~1 dB (§5.0); recording quality varies by Svarah speaker and bounds what each voice can sound like.
- **Telugu G2P is orthographic**: it transcribes the script; it does not model schwa reduction, loanword pronunciations (ఫ as /f/), or sandhi. It was not reviewed by a native speaker.
- **English abbreviations and acronyms are not expanded** ("Dr.", "IIIT"; stress sentence SE06).
- **Telugu clips that MFA could not align were dropped** (28); their transcripts were not inspected.
- **English validation shares speakers with training** (split by recording, §2.1).

8.3 Possible improvements (from the test sentences)

- **Voice and data selection by recording quality** (§5.0): select or weight training speakers by measured clarity, not only by amount of audio; the test-voice choice alone removed most English ASR errors.
- **Function words and number sequences** (E4): minimum durations for unstressed vowels, or a duration floor per phone class.
- **Early stopping / checkpoint selection** on validation loss (English ~7–10k steps).
- **Rare and long words**: a pronunciation lexicon for domain terms (E1 is now recognised, but relies on g2p_en).
- **Text normalisation**: abbreviations, acronyms (spelled letter by letter), ordinals in English dates ("12th November").
- **Vocoder fine-tuning** on each language's data (HiFi-GAN fine-tuned on ground-truth-aligned mels) — a smaller gain than first expected, since copy-synthesis already matches the recordings' spectrum (§5.0) estimated.
- **Telugu**: native-speaker review of the G2P rules; frame-level pitch with CWT (as in the paper) for richer intonation.

9. Reproducibility

Software (as run on Kaggle): Python 3.13.15, PyTorch 2.11.0 (CUDA 12.8 on T4; CPU build for data jobs), librosa, pyworld, tgt, g2p_en, MFA 3.4.3 (conda-forge, via micromamba), transformers 5.16.1 (evaluation only). Full package lists are in each job's `environment.txt`; `requirements.txt` lists the minimum versions. Data: Svarah and IndicTTS Telugu from Hugging Face at pinned revisions (`data/ingest.py`).

Configuration: `English/config.yaml`, `Telugu/config.yaml` (preprocess / model / train sections; step values as used), and the exact configs as run in `Results/<Lang>/train_config_used.yaml`.

Commands (data root `D`; Telugu shown, English analogous):

```
cd fastspeech2
python -m data.ingest --lang te --cache /tmp/hf --out D                  # needs a Hugging Face token
python -m data.select_split --manifest_dir D/manifests/te --train_hours 5 --val_hours 1
python data/mfa_prepare.py --lang te --ingest_dir D --corpus /tmp/corpus_te --dict_out
D/mfa_te/dictionary.txt
mfa train /tmp/corpus_te D/mfa_te/dictionary.txt D/mfa_te/model.zip --output_directory
D/preprocessed/te/TextGrid
#   English: mfa align /tmp/corpus_en D/mfa_en/dictionary.txt english_us_arpa D/preprocessed/en/TextGrid --
beam 100 --retry_beam 400
cd ..
python Telugu/preprocess.py --data_root D
python Telugu/train.py      --data_root D --max_minutes 480
python Telugu/inference.py  --data_root D --checkpoint Telugu/checkpoints/te_model.pth.tar --speaker
te_female --out_dir Results/Telugu
python fastspeech2/evaluate_val.py --config Telugu/config.yaml --data_root D \
       --checkpoint Telugu/checkpoints/te_model.pth.tar --speaker te_female --out_dir Evaluation/Telugu
```

Checkpoints: GitHub release v1.0 (`en_model.pth.tar`, `te_model.pth.tar`; SHA-256 in `<Lang>/checkpoints/README.md`). Inference alone needs only a checkpoint and the `stats.json` / `speakers.json` in `Results/<Lang>/`. **Kaggle jobs** (`kaggle/`) reproduce every stage, and each records the commit it ran. **Tests**: `pytest`.